Comparing Data Sets

Exam code: 0607

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Comparing data sets

How do I compare two data sets?

- You may be given **two** sets of data that relate to a context
- To **compare** data sets, you need to
  - compare their **averages**
    - Mode, median or mean
  - compare their **spreads**
    - Range

How do I write a conclusion when comparing two data sets?

- When comparing **averages** and **spreads**, you need to
  - compare **numbers**
  - **describe** what this means in **real life**
- **Copy** the **exact** wording from the question in your answer
- There should be **four** parts to your conclusion
  - For example:
    - "The **median** score of class A (45) is **higher** than the median score of class B (32)."
    - "This means class A **performed better** than class B in the test."
    - "The **range** of class A (5) is lower than the range of class B (12)."
    - "This means the scores in class A were **less spread out** than scores in class B."
  - Other good phrases for lower ranges include:
    - "scores are **closer together**"
    - "scores are **more consistent**"
    - there is **less variation** in the scores"

What restrictions are there when drawing conclusions?

- The data set may be **too small** to be truly representative
  - Measuring the heights of only 5 pupils in a whole school is not enough to talk about averages and spreads
- The data set may be **biased**
  - Measuring the heights of just the older year groups in a school will make the average appear too high

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

of average if it helps to strengthen their argument!

What else could I be asked?

- You may need to **choose** which, out of mode, median and mean, to compare
  - Check for **extreme values** (outliers) in the data
    - Avoid using the **mean** as it is affected by extreme values
- You may need to think from the **point of view** of another person
  - A teacher might not want a large spread of marks
    - It might show that they haven't taught the topic very well!
  - An examiner might want a large spread of marks
    - It makes it clearer when assigning grade boundaries, A, B, C, D, E, ...

Examiner Tips and Tricks

When comparing data sets in the exam, half the marks are for comparing the numbers and the other half are for saying what this means in real life.

Worked Example

Julie collects data showing the distances travelled by snails and slugs during a ten-minute interval. She records a summary of her findings, as shown in the table below.

	Median	Range
Snails	7.1 cm	3.1 cm
Slugs	9.7 cm	4.5 cm

Compare the distances travelled by snails and slugs during the ten-minute interval.

Answer:

Compare the numerical values of the median (an average)
Describe what this means in real life

Slugs have a higher median than snails (9.7 cm > 7.1 cm)
This suggests that, on average, slugs travel further than snails

Compare the numerical values of the range (the spread)
Describe what this means in real life

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